Unit: Unit 5: Linear Equations **Name:** ____________

Topic: Point-Slope Form, Slope-Intercept Form, Standard Form and Converting Forms **Date:** __________

Foundation (Jalapeno)

Q1. What is the point-slope form of a linear equation?

(A) $y - y_1 = m(x - x_1)$

(B) $y = mx + b$

(C) $ax + by = c$

(D) $x = my + b$

(E) $y = \frac{1}{m}x + b$

Q2. Which of the following is the slope-intercept form of a linear equation?

(A) $y - y_1 = m(x - x_1)$

(B) $y = mx + b$

(C) $ax + by = c$

(D) $x = my + b$

(E) $y = \frac{1}{m}x + b$

Q3. What is the standard form of the equation $2y = 5x - 3$?

(A) $5x - 2y = 3$

(B) $2y = 5x - 3$

(C) $5x + 2y = -3$

(D) $2x - 5y = -3$

(E) $2y = -5x + 3$

Q4. Convert the equation $y - 2 = 3(x - 1)$ to slope-intercept form.

(A) $y = 3x + 1$

(B) $y = 3x - 1$

(C) $y = -3x + 1$

(D) $y = -3x - 1$

(E) $y = \frac{1}{3}x + 1$

Q5. Graph the equation $y = 2x - 3$. What is the y-intercept?

(A) $(-3, 0)$

(B) $(0, -3)$

(C) $(3, 0)$

(D) $(0, 3)$

(E) $(1, 0)$

Q6. What is the slope of the line $3x - 2y = 5$?

(A) $\frac{3}{2}$

(B) $\frac{2}{3}$

(C) $-\frac{3}{2}$

(D) $-\frac{2}{3}$

(E) $\frac{5}{2}$

Q7. Convert the equation $y = \frac{2}{3}x + 1$ to point-slope form.

(A) $y - 1 = \frac{2}{3}(x - 0)$

(B) $y - 0 = \frac{2}{3}(x - 1)$

(C) $y - 1 = \frac{3}{2}(x - 0)$

(D) $y - 0 = \frac{3}{2}(x - 1)$

(E) $y - 1 = \frac{1}{2}(x - 0)$

Q8. What is the x-intercept of the line $2y = 5x - 3$?

(A) $(0, 0)$

(B) $(\frac{3}{5}, 0)$

(C) $(3, 0)$

(D) $(-3, 0)$

(E) $(-5, 0)$

Open Ended Questions (FRQ)

Q9. Graph the equation $y = -2x + 1$ and find the x- and y-intercepts.

Q10. Convert the equation $y = \frac{1}{2}x - 2$ to standard form and find the slope and y-intercept.

Chef's Tips

- Tip 1: Make sure to check the calculations.
- Tip 2: Use graph paper for graphing.
- Tip 3: Read the questions carefully.